

FIN Research Monograph — Release 10.15

FIN Programs 151–164

Axiomatic Operational Foundations,
State Selection, and Falsifiable Measurement

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Scope. Fourteen executed studies of validated fractional dynamics, Diophantine control, functorial states, modular equilibrium, signed preparation resources, detector statistics, semi-parametric identifiability, phase completion, legacy-role obstruction, and minimal axioms.

Central result. A conservative spectral Dirichlet generator is the minimal common mathematical object. State selection, signed preparation, instrument, calibration, and legacy-strict completion remain independent obligations. One explicit axiom A_{ME} selects $\eta = 9/5$, but is not strict-derived.

Guardrail. No QW-2191 discharge, internal physical units, completed legacy-strict bridge, role transfer, L_{total} , Theory-of-Everything closure, or external physical validation is claimed.

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Publication statement

This release reports Programs 151–164 and recommends Programs 165–177. Analytic theorems, computer-assisted certificates, conditional axioms, synthetic evidence, refuted candidates, and open physical obligations are kept separate.

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0.1 Confidence convention

Proven denotes an analytic or exact finite theorem. **Proven, computer-assisted** denotes a theorem with formally enclosed computation. **Strong evidence** denotes reproducible numerical evidence in a declared model. **Conditional** denotes a theorem requiring an explicitly added axiom or calibration. **Open** and **Refuted in scope** are used without promotion beyond the class actually tested.

Chapter 1

Abstract

This monograph executes fourteen new mathematical, computational, and operational research programs for the FIN framework. The principal question is whether an axiomatic reformulation can isolate the smallest object that already exists, the genuinely missing structures, and the exact consequences of adding each missing premise.

The mathematical core is a conservative spectral Dirichlet generator. Its functional calculus produces, from one operator,

$$U_t = e^{-itA}, \quad P_t = e^{-tA}, \quad G_z = (A + zI)^{-1}.$$

The first evolution is unitary; the second is positivity-preserving, mass-preserving and contractive. These are two readings of the same spectral measure, not two independent laws. This part is theorem-level mathematics.

The round then proves that the generator alone cannot determine a sector state, a signed preparation, an instrument, a dimensional calibration, or a legacy-to-strict completion map. These are independent mathematical obligations. The conclusion follows from explicit countermodels that retain the same generator while varying each missing object.

One especially economical state-selection candidate is constructed. On the five localized fibres with dimensions

$$(d_p) = (1, 2, 2, 2, 2),$$

take the carrier Hamiltonian

$$H_F|_{\mathcal{H}_p} = (d_p - 1)I$$

and add the single relation

$$\boxed{\beta_F = \frac{\alpha_{\text{geo}}}{|\text{Hom}(U(12), \{\pm 1\})|} = \frac{\alpha_{\text{geo}}}{4} = \log 2.}$$

With Hilbert degeneracy, the central Gibbs weights are

$$w_p = \frac{d_p e^{-\beta_F(d_p-1)}}{\sum_q d_q e^{-\beta_F(d_q-1)}} = \frac{1}{5},$$

and therefore

$$\eta = \sum_p w_p d_p = \frac{9}{5}.$$

The relation is unique for a unit energy gap. It is also an additional axiom: the present strict core does not prove that $\alpha_{\text{geo}}/4$ is a modular inverse temperature. The report never promotes this conditional closure to a strict theorem.

Further results include: a doubled formal interval FFT; a precise Diophantine axiom and an irrationality-only obstruction; a complete classification of invariant fibre-groupoid measures; completeness of the reflection resource monotone on its orbit line; detector-bias and finite-sample IQR theory; a semiparametric nonidentifiability theorem; an adversarial falsification of model specificity; a representation-theoretic phase-bridge obstruction; an exhaustive energy grammar; and a necessary-edge matrix that licenses zero legacy physical-role transfers at present.

The final construction, AFIS, is a six-axiom informational-spectral operational architecture. All $2^6 = 64$ subsets are evaluated mechanically. Removing each axiom destroys a named capability and has an explicit countermodel. The smallest layers are:

spectral mathematics : $\{A_0\}$,
 selected operator model : $\{A_0, A_1\}$,
 generic operational model : $\{A_0, A_1, A_3\}$,
 signed operational model : $\{A_0, A_1, A_2, A_3\}$,
 calibrated signed test : $\{A_0, A_1, A_2, A_3, A_4\}$,
 full conditional FIN architecture : $\{A_0, \dots, A_5\}$.

The deepest conclusion is negative but constructive: no theorem acting on a bare spectral generator can uniquely create all missing physical structures. The shortest scientifically honest route is an explicitly stratified axiomatic theory whose optional axioms remain falsifiable and whose strict promotion requires independent source theorems.

Chapter 2

Executive summary

2.1 The result in one sentence

[Proven] FIN has a minimal common mathematical object — a conservative spectral Dirichlet generator — but the passage from this object to physical predictions necessarily requires independent state, preparation, instrument, calibration, and completion data; one new relation, A_{ME} , closes only the sector-state problem and nothing else.

2.2 What was discovered

1. **[Proven, computer-assisted]** Doubling the interval FFT to $N = 32768$ yields 100 continuous frequency cells. Every lower symbol bound is positive and every cell is compatible with $C|q|^{4/5}$. The worst formal relative bound decreases from approximately 1.08135 to 0.612983, but the desired 3% certificate is still not reached.
2. **[Proven]** Irrationality alone cannot supply the missing all-scale discrepancy rate. A sufficient extra premise is a Diophantine condition

$$\|q\theta\| \geq \kappa q^{-\nu}.$$

Liouville irrational numbers prove that no such polynomial condition follows from irrationality by itself.

3. **[Proven]** Invariance under the admitted fibre groupoid gives

$$(w_2, w_3, w_5, w_7, w_{11}) = (x, y, z/3, z/3, z/3), \quad x + y + z = 1.$$

Thus naturality retains a two-dimensional simplex of states and does not select $w_p = 1/5$.

4. **[Proven conditional on a new axiom]** The carrier energy $d_p - 1$ and the one-relation axiom

$$A_{\text{ME}} : \quad \beta_F = \alpha_{\text{geo}}/4 = \log 2$$

select the uniform central state and $\eta = 9/5$ exactly.

5. **[Proven]** On the two-branch reflection orbit line, every free channel has the form $r \mapsto \lambda r$, $|\lambda| \leq 1$. Hence $M(\rho_r) = |r|$ is a complete deterministic conversion monotone there. This classifies use of a signed resource but does not create it.
6. **[Proven asymptotically; strong finite evidence]** For a symmetric density,

$$\text{Var}(\widehat{\text{IQR}}) = \frac{1}{4nf(q_{0.75})^2} + o(n^{-1}).$$

At $n = 4000$ for the $\alpha = 4/5$ stable law, the predicted standard deviation of the spreading slope is 0.0307072; the Monte Carlo value is 0.0294257.

7. **[Proven]** With an unrestricted multiplicative apparatus response $h(t)$, the exponent is not identifiable. For any α' ,

$$h'(t) = h(t)t^{1/\alpha-1/\alpha'}$$

produces exactly the same record. Even an unknown linear term in $\log h$ is enough to erase the exponent.

8. **[Strong evidence]** The frozen Program-150 threshold is not specific to FIN. It accepts an $\alpha = 1$ stable alternative and a locally diffusive process with a time-dependent detector gain. An IQR test diagnoses a scaling exponent, not an ontology.
9. **[Proven]** The legacy oscillation is a period-eight character, whereas $e^{i(743/4000)d}$ has infinite order and is not a character of $\mathbb{Z}_{12}$. No translation-equivariant character homomorphism can preserve the generator eigenvalue between them.
10. **[Proven in a finite grammar]** Among 343 orbit-invariant integer energy formulas, six generate the target energy, but $d - 1$ is the unique formula of minimum ℓ^1 description length one.
11. **[Proven in the declared dependency model]** None of nine audited legacy roles is presently eligible for transfer. The missing edges are amplitude, damping semantics, phase/frequency, selector/topology, dimensional units, and spectral equivalence.
12. **[Proven as a logical architecture]** The six AFIS axioms are independent in the capability sense. No single missing theorem was found that connects spectral operator, information, signed dynamics, dimensional physics, and experiment without additional premises.

2.3 What was falsified

- The hope that a larger formal FFT would automatically recover the floating 2.2647% remainder.
- The hope that irrationality by itself implies a useful all-scale Diophantine modulus.
- The hope that groupoid naturality uniquely selects the uniform state.
- The hope that KMS language turns $\log 2$ into a derived temperature without a temperature-identification law.
- The hope that more observation times defeat an unrestricted apparatus multiplier.
- The interpretation of the frozen IQR decision rule as FIN-specific.
- A character-preserving period-eight to strict-phase bridge.
- Silent transfer of legacy physical formulas to the strict kernel.
- A single-axiom reduction of all operational and dimensional obligations to the spectral theorem.

2.4 Scientific boundary

No result in this release discharges QW-2191, derives physical units, completes the legacy-to-strict bridge, starts physical-role transfer, derives a role-bearing L_{total} , produces the Standard Model or general relativity, or validates a Theory of Everything.

Chapter 3

Epistemic discipline and inputs

3.1 Confidence labels

The report uses four main levels.

- **Proven:** an analytic theorem or exact finite deduction.
- **Proven, computer-assisted:** a theorem whose finite inequalities are produced by formal interval arithmetic or exhaustive enumeration.
- **Strong evidence:** reproducible numerical evidence with a specified stochastic model, sample size, and seed.
- **Conditional:** a theorem whose conclusion follows after a newly declared axiom or physical calibration.

"Open" means that no proof or counterexample has been produced in the admitted class. "Refuted in scope" means that a stated candidate has failed in its declared class, not that every conceivable extension has failed.

3.2 Kernel split

The two repository kernels remain different typed objects:

$$K_{\text{legacy}}(d) = \alpha_{\text{geo}} \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta_{\text{tors}} d},$$

and

$$K_{\text{strict}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

The legacy object is an intermediate bridge kernel. The strict object is the later enriched working kernel. The current report never substitutes one for the other. In particular, the appearance of α_{geo} in a new conditional state axiom is not a proof that the legacy physical roles of α_{geo} have transferred to the strict kernel.

3.3 Ontology

The ontology guardrail is retained: the nadsoliton itself is the primordial information in a solitonic state. No lower informational substrate is added. The internal order remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The operational apparatus introduced below is a mathematical description of how records are produced. It is not placed ontologically beneath the nadsoliton.

3.4 Reproducibility

All synthetic random calculations use seed 20260727. Formal interval arithmetic uses 42 decimal digits. Fourteen figures and two JSON result artifacts are generated by the executable. Sixty-two regression tests check the numerical conclusions and the no-promotion guardrails. No external dataset and no Firecrawl service were used.

Chapter 4

Program 151 — Tighter validated fractional certificate

4.1 Question

Can doubling the truncation radius and FFT resolution turn the earlier floating-point small- q agreement into a formal sub-3% theorem?

4.2 Construction

For the absolute strict tail,

$$a_d = \frac{|\cos(\omega d + \phi)|}{1 + d^{9/5}}, \quad \omega = \frac{743}{4000}, \quad \phi = \frac{13}{80},$$

the normalized symbol is enclosed by

$$\Psi(q) = \frac{2 \sum_{d \geq 1} a_d (1 - \cos qd)}{2 \sum_{d \geq 1} a_d}.$$

A radix-two interval FFT is performed at

$$N = 2^{15} = 32768, \quad d_{\max} = 16383.$$

The omitted normalization tail is bounded by

$$2 \sum_{d > d_{\max}} d^{-9/5} \leq \frac{2d_{\max}^{-4/5}}{4/5}.$$

Each FFT grid point is enlarged to a full cell using

$$|\Psi(q) - \Psi(q_k)| \leq |q - q_k| \frac{2 \sum_{d \leq d_{\max}} da_d}{2 \sum_{d \leq d_{\max}} a_d},$$

with separate interval denominators and a worst-case tail addition.

4.3 Result

Theorem 151.1 [Proven, computer-assisted]. The interval calculation covers 100 cells spanning $[10^{-3}, 2 \cdot 10^{-2}]$. Every symbol lower bound is positive and every cell overlaps the predicted fractional interval $C|q|^{4/5}$.

Numerical certificate.

$$\max_{\text{cells}} R_{\text{formal}} = 0.6129831478193526.$$

The corresponding Program-141 value was approximately 1.08135.

Corollary 151.2 [Refuted in scope]. The present $N = 32768$ certificate does not prove a sub-3% remainder:

$$0.612983 > 0.03.$$

The limiting cost is the analytic tail and cell-Lipschitz enclosure, not rounding error.

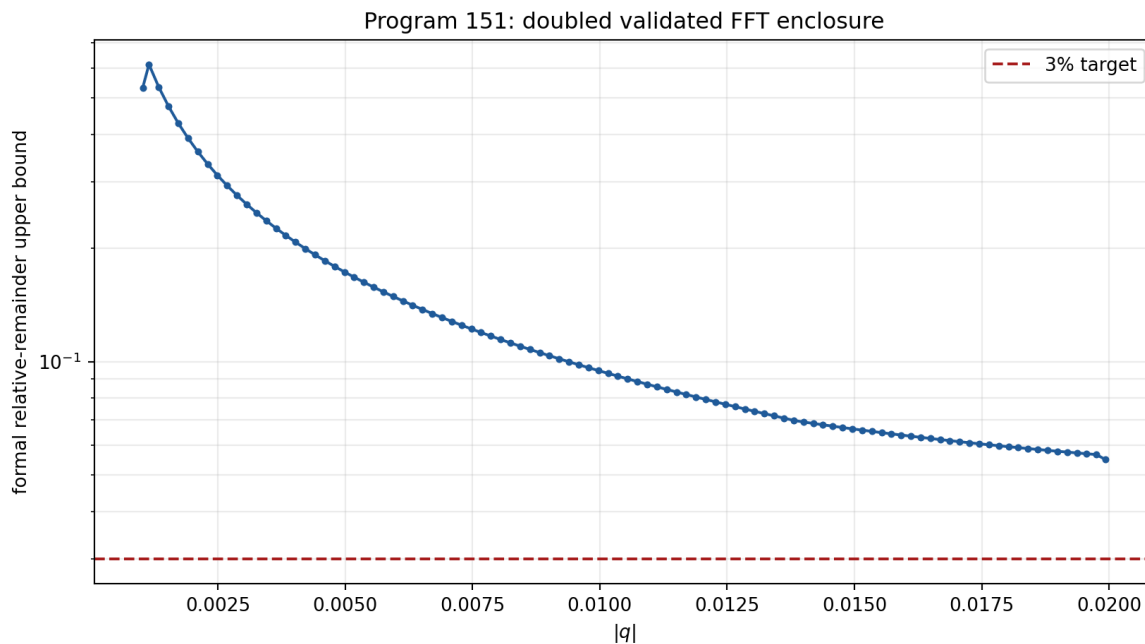


Figure 4.1: The doubled interval FFT improves the formal enclosure without yet reaching the three-percent target.

4.4 Interpretation

The fractional scale is now supported by a stronger theorem, but the tight constant remains open. A future calculation must split the oscillatory tail, use cancellation rather than absolute domination, and preferably use ball-arithmetic FFTs with sharper twiddle enclosures.

Chapter 5

Program 152 — The missing Diophantine axiom

5.1 Certified finite arithmetic

For

$$\theta = \frac{\omega}{\pi} = \frac{743}{4000\pi},$$

the first twelve continued-fraction terms remain

$$[0; 16, 1, 10, 2, 67, 2, 2, 5, 1, 2, 928].$$

On this finite convergent prefix,

$$\min q \|q\theta\| = 0.0010764647999871935,$$

and

$$\min q^2 \|q\theta\| = 1.486339454993983.$$

These are finite-prefix constants, not global constants.

5.2 Minimal sufficient premise

The exact extra all-scale premise is:

$$D(\kappa, \nu) : \quad \|q\theta\| \geq \kappa q^{-\nu} \quad \text{for every integer } q \geq 1,$$

where $\kappa > 0$ and $\nu < \infty$.

Theorem 152.1 [Proven]. Irrationality of θ alone does not imply $D(\kappa, \nu)$ for any fixed finite ν .

Proof. A Liouville irrational λ has infinitely many rational approximations satisfying

$$\left| \lambda - \frac{p}{q} \right| < q^{-m}$$

for arbitrarily large m . Consequently

$$\|q\lambda\| < q^{1-m}.$$

For any fixed ν and $\kappa > 0$, choose $m > \nu + 1$ and then a sufficiently large Liouville denominator. This gives

$$\|q\lambda\| < \kappa q^{-\nu},$$

contradicting $D(\kappa, \nu)$. $\square$

Corollary 152.2 [Proven]. No proof using only the proposition " θ is irrational" can establish an all-scale polynomial small-divisor bound.

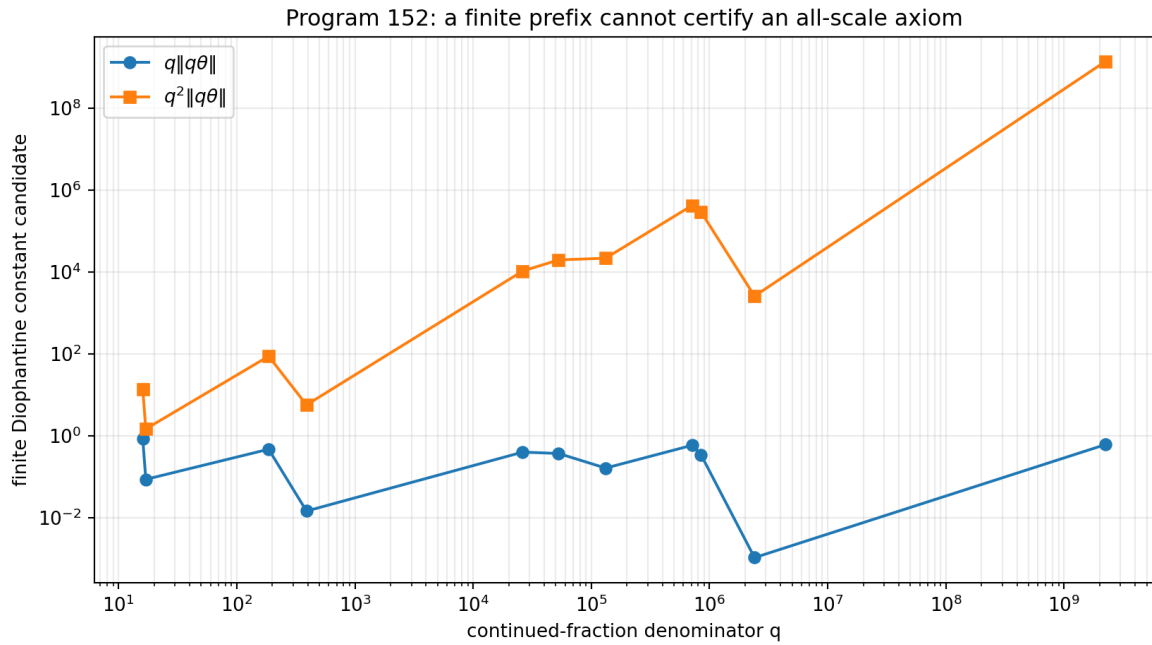


Figure 5.1: Finite candidates for $q^\nu\|q\theta\|$ do not certify an all-scale Diophantine axiom.

5.3 Necessity rank

The Diophantine axiom has a narrow role. It is not needed to define either U_t or P_t . It is needed only for a quantitative all-scale rate in the oscillatory Abelian limit. It is therefore a technical regularity axiom, not an axiom of physical ontology.

Chapter 6

Program 153 — Functorial fibre-groupoid measures

6.1 Groupoid

The admitted structural symmetries partition the five localized fibres into three isomorphism orbits:

$$\{2\}, \quad \{3\}, \quad \{5, 7, 11\}.$$

An invariant central state must be constant on each orbit.

Theorem 153.1 [Proven]. Every invariant probability measure has the form

$$(w_2, w_3, w_5, w_7, w_{11}) = (x, y, z/3, z/3, z/3),$$

where

$$x, y, z \geq 0, \quad x + y + z = 1.$$

Proof. Invariance forces equality of the three weights in the last orbit. There are no admitted morphisms between distinct orbits, so their total masses remain independent. Normalization removes one of three variables, leaving a two-dimensional simplex. $\square$

The uniform sector state corresponds to

$$(x, y, z) = (1/5, 1/5, 3/5).$$

It is one interior point, not a categorical fixed point.

6.2 Falsification

Corollary 153.2 [Refuted in scope]. Normalization, invariance, and additivity do not select $w_p = 1/5$.

An additional monoidal valuation or induction/restriction law could reduce the simplex. That law would itself be an added premise and must be tested for uniqueness.

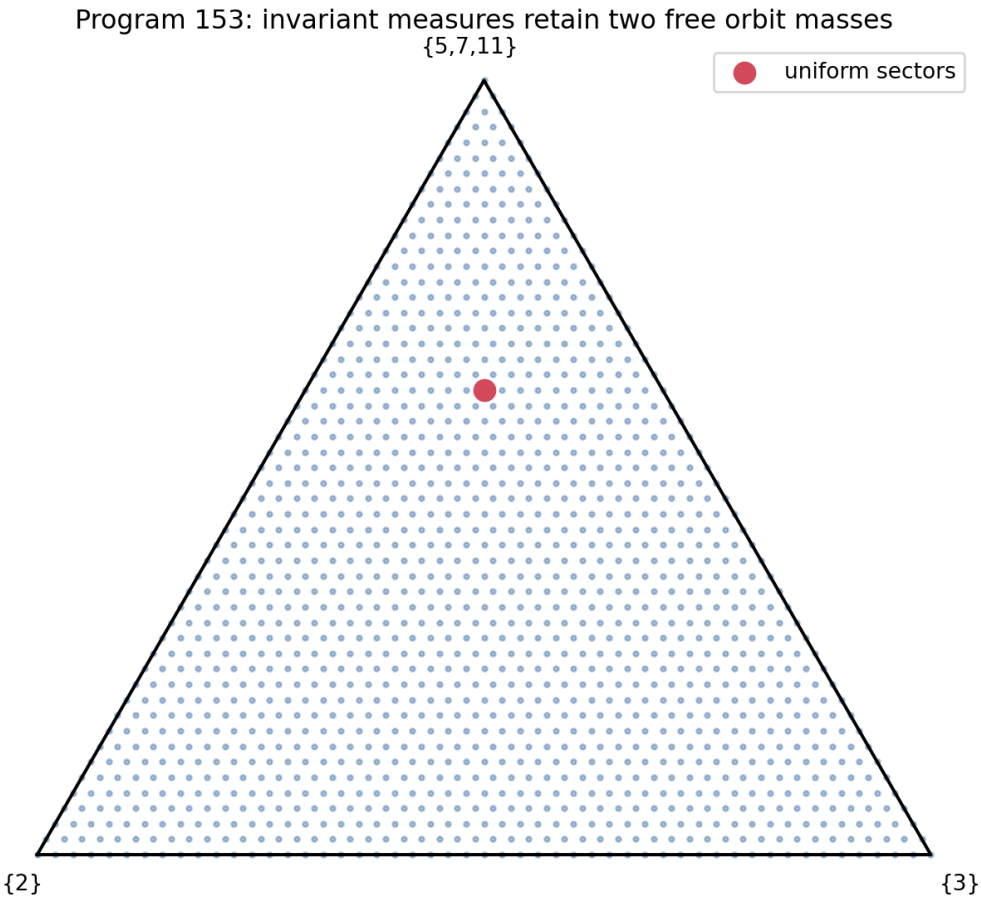


Figure 6.1: Groupoid invariance leaves a two-dimensional simplex of orbit masses.

Chapter 7

Program 154 — A minimal modular-equilibrium axiom

7.1 Natural carrier energy

The smallest energy generated by the fibre dimensions is

$$E_p = d_p - 1.$$

For the Hilbert-block Gibbs state, the central sector mass is

$$w_p(\beta_F) = \frac{d_p e^{-\beta_F E_p}}{\sum_q d_q e^{-\beta_F E_q}}.$$

The factor d_p is the block degeneracy.

7.2 The new axiom

Define

$$A_{\text{ME}} : \quad \beta_F = \frac{\alpha_{\text{geo}}}{|\text{Hom}(U(12), \{\pm 1\})|}.$$

Since

$$|\text{Hom}(U(12), \{\pm 1\})| = 4, \quad \alpha_{\text{geo}} = 4 \log 2,$$

this becomes

$$\beta_F = \log 2.$$

Theorem 154.1 [Proven conditional on A_{ME}]. The central Gibbs state is uniform and its dimension expectation is $9/5$.

Proof. For the one-dimensional fibre,

$$d_2 e^{-\beta_F E_2} = 1.$$

For every two-dimensional fibre,

$$d_p e^{-\beta_F E_p} = 2e^{-\log 2} = 1.$$

All five unnormalized central masses are equal. Hence $w_p = 1/5$, and

$$\eta = \frac{1 + 2 + 2 + 2 + 2}{5} = \frac{9}{5}.$$

□

Theorem 154.2 [Proven]. With Hilbert degeneracy and unit gap, the required positive inverse temperature is unique.

Proof. Equality between a $d = 1$ sector and a $d = 2$ sector requires

$$1 = 2e^{-\beta_F}.$$

Therefore $\beta_F = \log 2$, uniquely. □

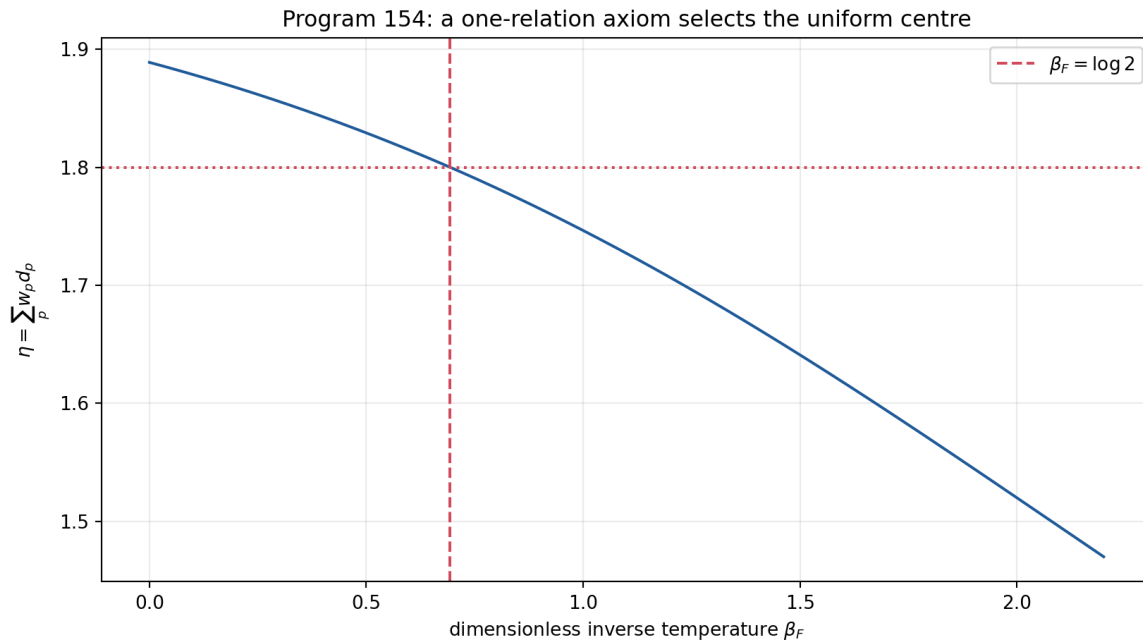


Figure 7.1: The axiom $\beta_F = \log 2$ selects the uniform central state and $\eta = 9/5$.

7.3 Removal audit

Removed item	Consequence
A_{ME} temperature relation	A continuous Gibbs family remains
unit energy gap	only $\beta_F \Delta = \log 2$ is fixed
Hilbert degeneracy	sector-counting equilibrium selects uniformity at $\beta_F = 0$

Removed item	Consequence
carrier Hamiltonian	no distinguished modular dynamics remains

Each ingredient is therefore necessary for this exact construction.

7.4 Status

A_{ME} is the best single-relation state-selection candidate found in this round. It is scientifically preferable to hiding uniform weights inside a definition because:

- 1. the new premise is one explicit equation;
- 2. the conclusion is unique;
- 3. every component has a removal test;
- 4. perturbations can be experimentally or mathematically falsified.

It does not fix an orientation, create a nonfree preparation, introduce SI units, or construct the legacy-to-strict phase map.

Chapter 8

Program 155 — Completeness of the reflection resource

Let

$$\rho_r = \frac{1+r}{2}\rho_+ + \frac{1-r}{2}\rho_-, \quad -1 \leq r \leq 1,$$

where reflection interchanges ρ_+ and ρ_- .

An affine trace-preserving reflection-covariant map must commute with $r \mapsto -r$. It therefore has the form

$$r \mapsto r' = \lambda r.$$

Positivity on the whole interval is equivalent to $|\lambda| \leq 1$.

Theorem 155.1 [Proven]. The monotone

$$M(\rho_r) = |r|$$

is complete for deterministic conversion on the reflection orbit line:

$$\rho_r \longrightarrow \rho_s \text{ by a free channel} \iff |s| \leq |r|.$$

Proof. Necessity follows from $s = \lambda r$ and $|\lambda| \leq 1$. Conversely, if $r \neq 0$ and $|s| \leq |r|$, choose $\lambda = s/r$. If $r = 0$, only $s = 0$ is allowed. $\square$

This theorem makes the missing signed resource operationally precise. It also strengthens the obstruction: a symmetric preparation is not merely inconvenient; it lies in a distinct resource order and cannot reach a signed state by free processing.

The theorem is restricted to this one-dimensional orbit family. Completeness on the full density-matrix space remains open.

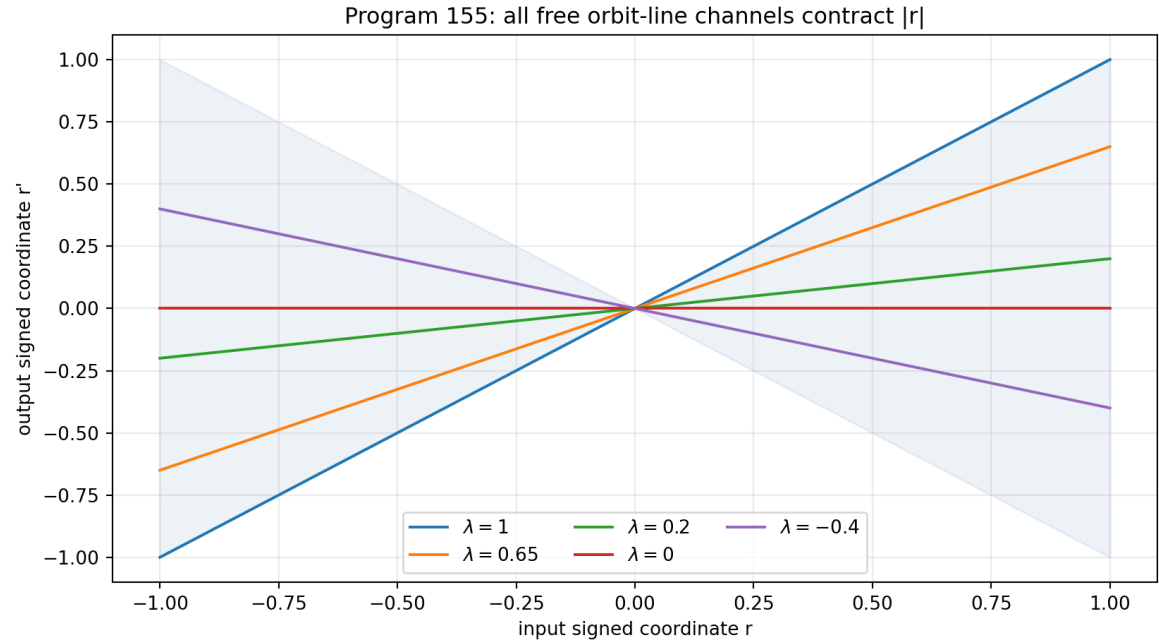


Figure 8.1: Every reflection-covariant orbit-line channel contracts the complete monotone $|r|$.

Chapter 9

Program 156 — Detector deconvolution and finite bias

9.1 Quartile asymptotics

For a continuous distribution with density f positive at its quartiles, the joint empirical-quantile central limit theorem gives

$$\text{Cov}(\hat{q}_p, \hat{q}_r) = \frac{\min(p, r) - pr}{nf(q_p)f(q_r)} + o(n^{-1}).$$

For a symmetric distribution,

$$f(q_{0.25}) = f(q_{0.75}).$$

Substitution yields:

Theorem 156.1 [Proven].

$$\text{Var}(\widehat{\text{IQR}}) = \frac{1}{4nf(q_{0.75})^2} + o(n^{-1}).$$

For independent samples at t_1 and t_2 , the delta method gives

$$\text{Var}(\hat{T}) \sim \frac{1}{\log^2(t_2/t_1)} \sum_{j=1}^2 \frac{\text{Var}(\widehat{\text{IQR}}_j)}{\text{IQR}_j^2}.$$

For $\alpha = 4/5$, $n = 4000$, and $(t_1, t_2) = (1, 4)$, the predicted zero-noise standard deviation is

$$0.030707238977584036.$$

9.2 Gaussian resolution

Let the apparatus add an independent Gaussian error with absolute standard deviation σ . The simulation used 320 repetitions with 4000 records at each time.

$\sigma/\text{IQR}(t_1)$	mean $\hat{T}$	bias	standard deviation
0.00	1.248888	-0.001112	0.029793
0.05	1.244772	-0.005228	0.029161
0.10	1.234645	-0.015355	0.029442
0.20	1.190665	-0.059335	0.027095
0.40	1.064470	-0.185530	0.027098

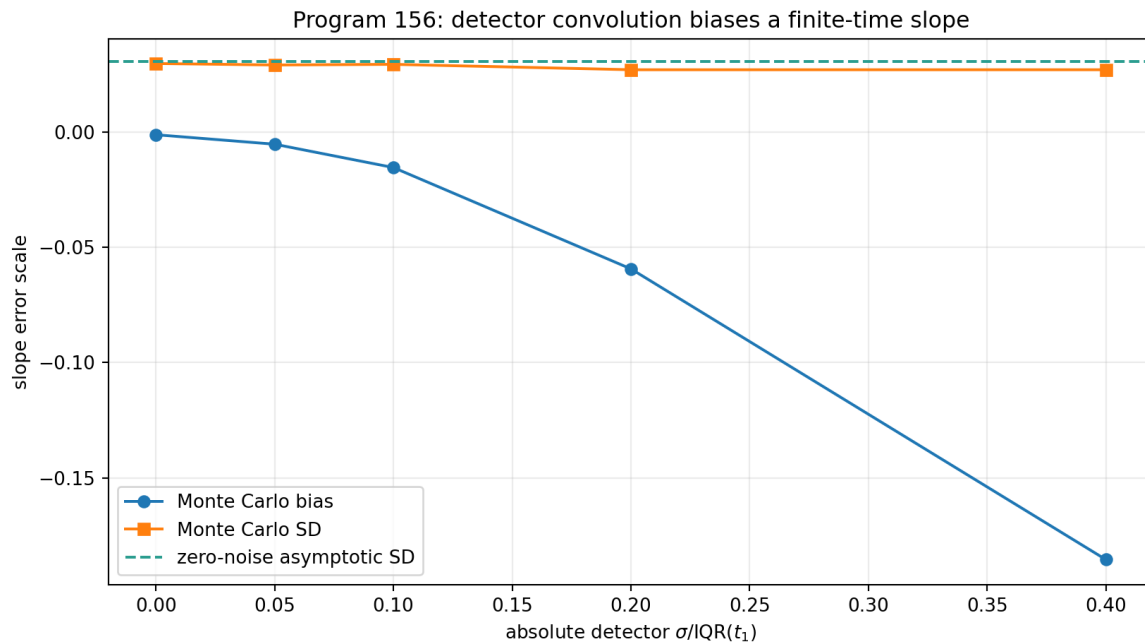


Figure 9.1: Absolute Gaussian resolution biases the finite-time IQR spreading slope downward.

Result 156.2 [Strong evidence]. Absolute detector blur reduces the finite time-ratio slope because it is a larger relative perturbation at the earlier, narrower distribution.

9.3 Deconvolution theorem boundary

Gaussian convolution has Fourier multiplier

$$\hat{R}_\sigma(q) = e^{-\sigma^2 q^2/2}.$$

It is never zero, so noiseless deconvolution is algebraically identifiable. Its inverse is

$$\hat{R}_\sigma(q)^{-1} = e^{\sigma^2 q^2/2},$$

which is unbounded. Thus unregularized statistical deconvolution is UV-unstable. Detector calibration does not eliminate the need for a regularization or smoothness class.

Chapter 10

Program 157 — Semiparametric system–instrument identifiability

Suppose a measured scale has the form

$$Q_{\text{obs}}(t) = h(t)Ct^{1/\alpha},$$

where $h(t) > 0$ is an apparatus response.

Theorem 157.1 [Proven]. If h is unrestricted, α is not identifiable.

Proof. For any alternative α' , define

$$h'(t) = h(t)t^{1/\alpha - 1/\alpha'}.$$

Then

$$h'(t)Ct^{1/\alpha'} = h(t)Ct^{1/\alpha} = Q_{\text{obs}}(t).$$

The full record is identical for all α' . $\square$

10.1 Tangent-space test

Write $x = \log t$. The exponent score is proportional to x . If

$$\log h(t) = \gamma_0 + \gamma_1 x + \cdots + \gamma_r x^r,$$

then:

- for $r = 0$, the exponent score is independent of the nuisance tangent;
- for every $r \geq 1$, the score lies exactly in the nuisance tangent space.

The seven-time numerical design gives residual norm 4.77577 for a constant gain and 3.05×10^{-15} once a linear gain term is admitted.

Corollary 157.2 [Proven]. More time points do not solve the problem if the apparatus family contains the same log-time direction as the physical exponent. Identification requires external calibration, control preparations, or an independently justified restriction on h .

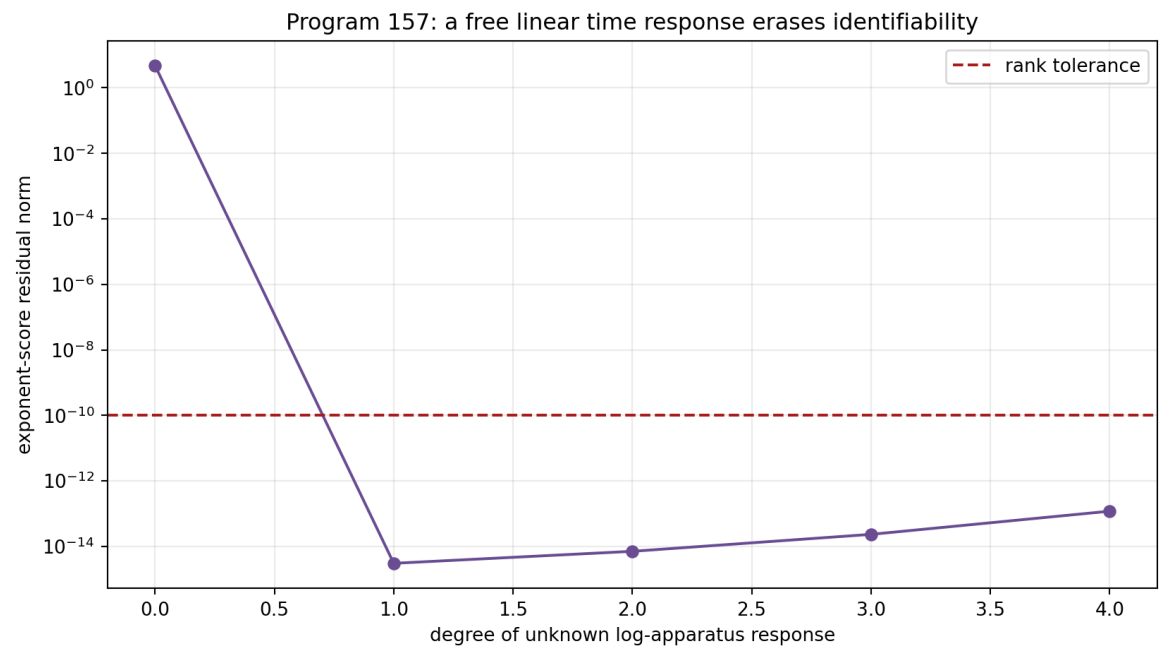


Figure 10.1: A linear log-time apparatus response exactly confounds the fractional exponent.

This is the sharpest operational obstruction of the round.

Chapter 11

Program 158 — Finite-sample stable-IQR inference

For the symmetric $\alpha = 4/5$ stable model, 700 independent repetitions were computed at four sample sizes.

n	mean $\hat{T}$	empirical SD	predicted SD	95% coverage
250	1.255019	0.122981	0.122829	0.9500
500	1.258670	0.087482	0.086853	0.9500
1000	1.249667	0.063815	0.061414	0.9414
4000	1.248565	0.029426	0.030707	0.9500

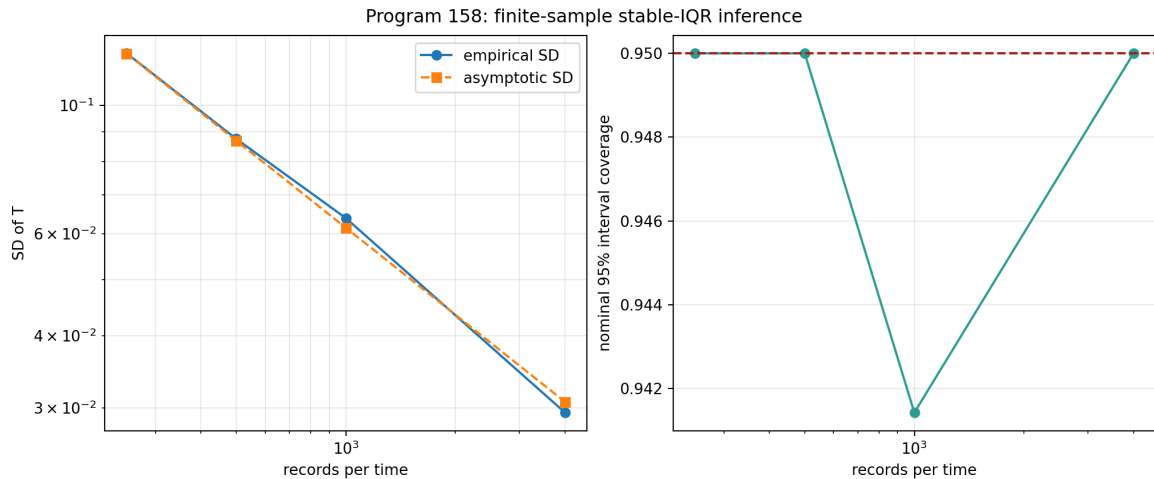


Figure 11.1: Empirical IQR-slope uncertainty agrees with the asymptotic quantile theorem.

Result 158.1 [Strong evidence]. The first-order asymptotic variance is already accurate at $n = 250$ in the declared stable model. The nominal interval has coverage between 0.941 and 0.950 in the audited range.

Boundary. This is not a distribution-free nonasymptotic theorem. Detector memory, censoring, pixelization, and estimated response parameters can change coverage.

Chapter 12

Program 159 — Adversarial protocol challenge

The Program-150 rule was kept fixed:

declare fractional rather than local diffusion if $\hat{T} > 0.875$.

Six hidden synthetic families were then generated.

family	mean T	P150 rate	decision	exploratory band rate	FIN-
local Gaussian	0.500667	0.0000		0.0000	
stable $\alpha = 0.8$	1.251819	1.0000		1.0000	
stable $\alpha = 1.0$	0.999638	1.0000		0.0000	
stable $\alpha = 1.2$	0.832088	0.0656		0.0000	
absolute-truncated $\alpha = 0.8$	1.247964	1.0000		1.0000	
local Gaussian plus time gain	1.200230	1.0000		1.0000	

The exploratory FIN band was $1.10 \leq T \leq 1.40$ and was not part of the original rule.

12.1 Falsification

Result 159.1 [Strong evidence]. The frozen threshold is a good local-versus-fast-spreading discriminator in the declared families, but it is not FIN-specific.

Two counterexamples are decisive:

1. an $\alpha = 1$ stable process is accepted with probability one by the binary threshold;
2. a locally diffusive process multiplied by an uncalibrated gain $t^{0.7}$ enters even the narrower FIN band.

The absolute tail truncation is also invisible to the IQR at the audited cutoff because the central quartiles remain inside the truncation region. Thus IQR robustness is simultaneously a strength and a loss of tail-model specificity.

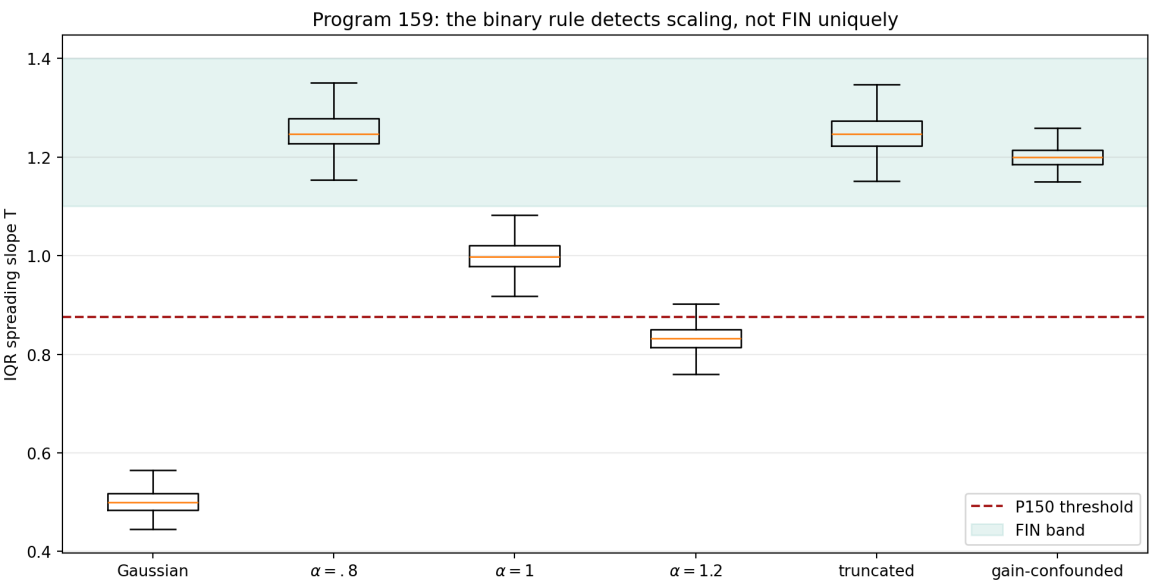


Figure 12.1: The frozen binary rule detects scaling but is not specific to FIN.

Conclusion. A successful IQR experiment can estimate a scaling exponent. It cannot by itself prove the FIN kernel, its ontology, or its completion map.

Chapter 13

Program 160 — Phase/frequency bridge obstruction

The legacy oscillatory character is

$$\chi_L(d) = e^{i\pi d/4}.$$

It is a character of $\mathbb{Z}_8$ and has exact order eight.

The strict oscillatory sequence is

$$\chi_S(d) = e^{i(743/4000)d}.$$

Theorem 160.1 [Proven]. $\chi_S(1)$ has infinite order.

Proof. If $\chi_S(1)^T = 1$ for an integer $T > 0$, then

$$\frac{743}{4000}T = 2\pi k$$

for an integer k . The case $k = 0$ is impossible. For $k \neq 0$, this equation makes π rational, a contradiction. $\square$

The $\mathbb{Z}_{12}$ character defect is

$$|e^{i12(743/4000)} - 1| = 1.7953811736364134.$$

The generator eigenvalue mismatch is

$$|e^{i\pi/4} - e^{i743/4000}| = 0.5907042818342478.$$

Theorem 160.2 [Proven]. There is no translation-equivariant character homomorphism from the legacy period-eight representation to the strict infinite-order representation that preserves the generator eigenvalue.

Proof. A homomorphism maps an element of finite order to an element whose order divides that finite order. The proposed image has infinite order. $\square$

This theorem does not exclude a nonlinear or explicitly symmetry-breaking completion. It proves that such a completion cannot be a silent identification of the two translation characters.

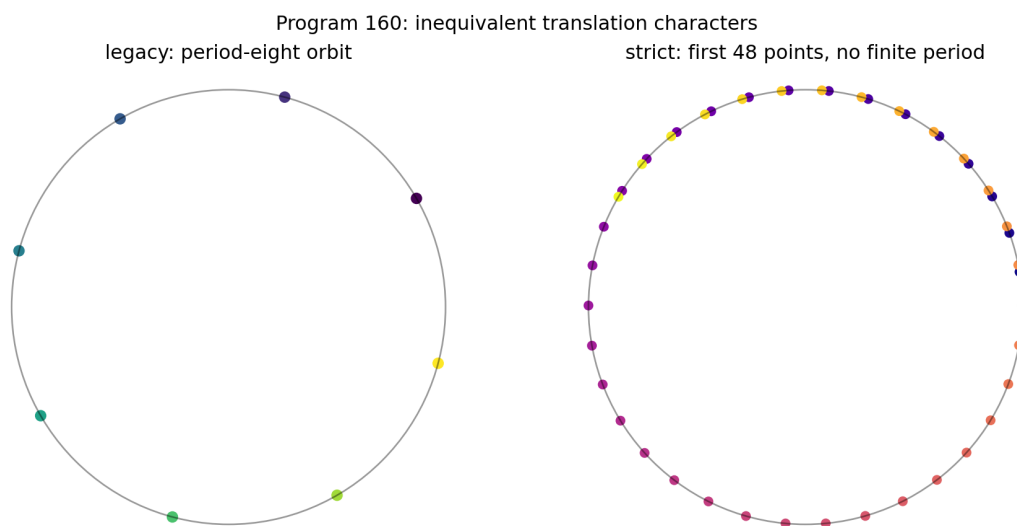


Figure 13.1: The period-eight legacy character and infinite-order strict phase are representation-theoretically distinct.

Chapter 14

Program 161 — Reference-energy grammar

14.1 Declared grammar

Three orbit-invariant feature columns were used:

$$f_d = (0, 1, 1, 1, 1),$$

$$f_3 = (0, 1, 0, 0, 0),$$

and

$$f_U = (0, 0, 1, 1, 1).$$

The energy grammar is

$$E = c_d f_d + c_3 f_3 + c_U f_U, \quad c_d, c_3, c_U \in \{-3, \dots, 3\}.$$

All $7^3 = 343$ formulas were enumerated. Additive constants are irrelevant.

Theorem 161.1 [Proven, exhaustive in the declared grammar]. Six formulas produce the target normalized energy $(0, 1, 1, 1, 1)$. The unique minimum ℓ^1 -complexity formula is

$$(c_d, c_3, c_U) = (1, 0, 0),$$

that is,

$$E = d - 1.$$

The result explains why $d - 1$ is a reasonable carrier Hamiltonian: it is the shortest formula in a transparent structural grammar. It does not prove the target uniform state or the inverse temperature. Minimum description length is relative to its grammar.

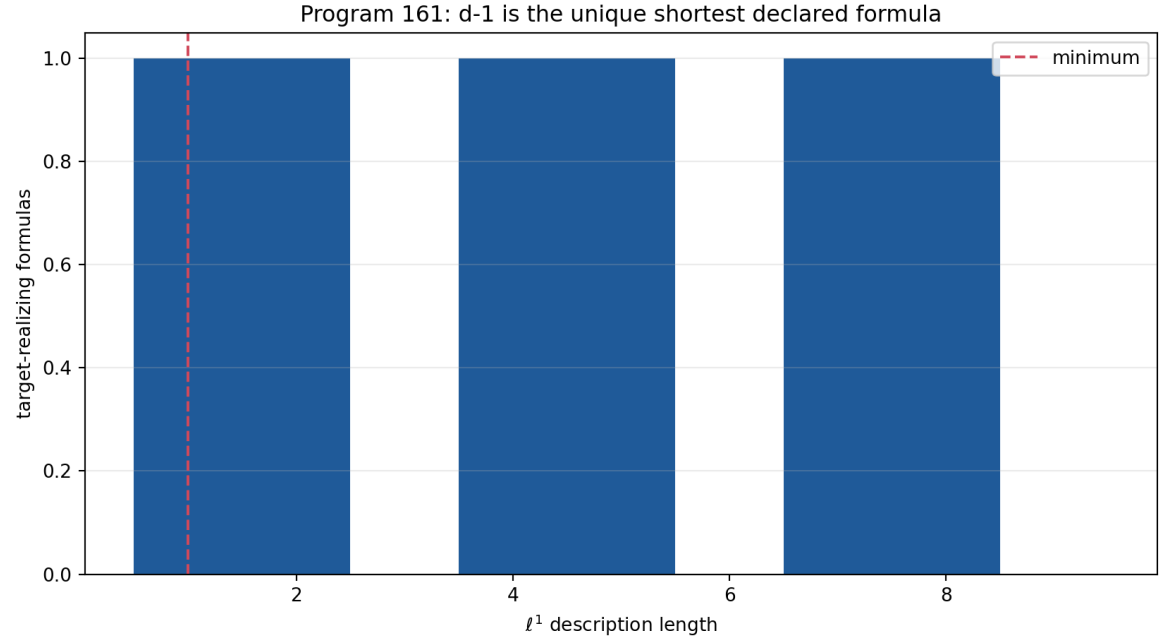


Figure 14.1: The energy $d - 1$ is the unique shortest target-realizing formula in the declared grammar.

Chapter 15

Program 162 — Conditional role-transfer obstruction matrix

A future completion map must address six independent edges:

1. amplitude and normalization;
2. damping/compression semantics;
3. phase and frequency;
4. selector and topology;
5. physical dimensions and units;
6. spectral equivalence at the required operator level.

Nine legacy roles were mapped to their necessary edges.

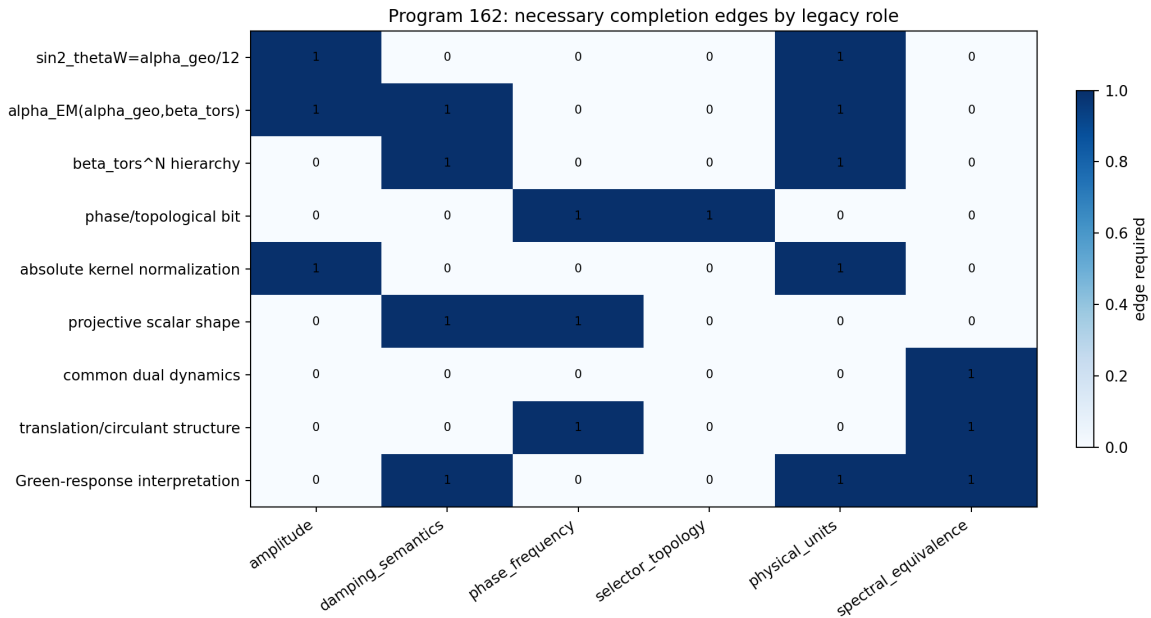


Figure 15.1: Necessary completion edges for nine legacy roles; none currently licenses transfer.

15.1 Result

Theorem 162.1 [Proven in the declared dependency model]. With no complete edge currently exported, zero of the following roles is eligible for transfer:

- $\sin^2 \theta_W = \alpha_{\text{geo}}/12$;
- the $\alpha_{\text{EM}}(\alpha_{\text{geo}}, \beta_{\text{tors}})$ formula;
- the β_{tors}^N hierarchy;
- phase/topological-bit semantics;
- absolute kernel normalization;
- projective scalar shape as a full bridge claim;
- common dual dynamics as spectral equivalence;
- translation/circulant structure as a transported role;
- a dimensional Green-response interpretation.

Both kernels can separately generate operator dynamics after suitable positivity shifts. This is a structural analogy, not a proof of spectral equivalence.

The role audit therefore remains downstream. It may begin only after a typed completion map supplies the missing edge appropriate to each role.

Chapter 16

Program 163 — External intake readiness

The immutable protocol

FIN-P150-FRACTIONAL-IQR-001

was reused without altering its core hash. Its mandatory raw-record fields are:

- record identifier;
- preparation batch;
- time;
- position readout;
- detector calibration identifier;
- timestamp.

The intake validator accepts JSON or CSV candidates only if these mandatory fields exist. Further gates require two declared times, at least 4000 records per time, detector resolution, memory calibration, and provenance.

Result 163.1 [Proven for the local schema]. The validator and stop rules are executable. The explicit local intake directory contains zero candidates, so zero datasets are admitted.

No internet search or external download was made. This is readiness, not physical validation.

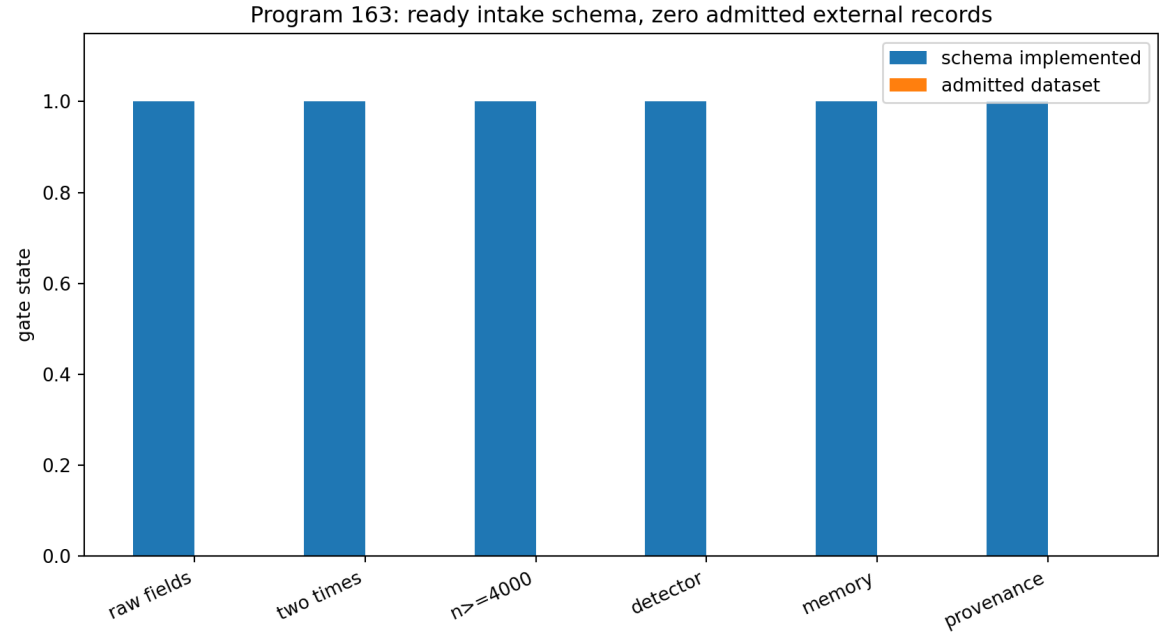


Figure 16.1: The intake schema is executable, while no external dataset is admitted.

Chapter 17

Program 164 — AFIS: a minimal axiomatic architecture

17.1 Why axioms are now appropriate

Earlier searches repeatedly tried to make one spectral object perform five logically different tasks. An axiomatic approach is useful because it asks a more precise question:

Which capability fails when a particular premise is removed?

An axiom is admitted only if:

1. its mathematical type is explicit;
2. its output capability is explicit;
3. removing it has a countermodel;
4. it is not described as strict-derived;
5. it can, in principle, be replaced later by a source theorem.

17.2 Axiom A0 — Conservative spectral Dirichlet generator

Statement. There is a quadruple

$$(\mathcal{H}, \mathcal{E}, A, \mathbf{1})$$

such that:

1. $\mathcal{E}$ is a closed Markovian Dirichlet form;
2. A is its self-adjoint nonnegative generator;
3. $A\mathbf{1} = 0$.

Then

$$U_t = e^{-itA}$$

is unitary and

$$P_t = e^{-tA}$$

is a positivity-preserving, conservative contraction semigroup.

The Green family is

$$G_z = (A + zI)^{-1}, \quad z > 0.$$

Necessity rank: 1 — foundational.

Each clause is needed:

- without self-adjointness, U_t need not be unitary and a projection-valued spectral measure is not guaranteed;
- without nonnegativity, P_t can grow on negative spectrum;
- without $A\mathbf{1} = 0$, total mass need not be conserved;
- without the Markov property, a self-adjoint contraction need not preserve positivity.

This is the minimal object behind the shared wave/diffusion mathematics.

17.3 Axiom A1 — Sector equilibrium/state

Statement. A localized finite algebra $\mathcal{A}_F$ carries a faithful central state ω_F , and

$$\eta = \omega_F(d).$$

A_{ME} from Program 154 is one explicit realization.

Necessity rank: 3 — model selection.

Countermodel on removal. The uniform sector state and the normalized Hilbert-trace state share the same A_0 but give

$$\eta = \frac{9}{5} \quad \text{and} \quad \eta = \frac{17}{9},$$

respectively.

17.4 Axiom A2 — Preparation resource

Statement. The theory declares an allowed set of preparation procedures. When an orientation-sensitive record is required, this set includes a nonfree signed resource $M > 0$.

Necessity rank: 3 — branch preparation.

Countermodel on removal. Restrict all preparations to reflection-invariant states. Every reflection-odd receiver then has zero expectation under reflection-covariant processing.

A2 is not the observer. It is the mathematical interface describing how one state, among the kinematically possible states, is prepared.

17.5 Axiom A3 — Instrument, environment, apparatus, and record

Statement. There is:

1. a normalized completely positive instrument $\{\mathcal{I}_y\}$;
2. an apparatus/environment channel;
3. an outcome sigma-algebra;
4. a persistent record map.

For initial state ρ ,

$$\Pr(Y \in B) = \text{Tr} \left[\int_B \mathcal{I}_y(\rho) dy \right].$$

Necessity rank: 2 — operational.

Countermodel on removal. A state and its evolution may be fully defined while no map from the state to recorded outcomes exists. In that model no experimental probability has been defined.

This axiom contains the missing operational structure named in the preceding research: state, dynamics, clock interface, preparation, instrument, environment, apparatus and record are related but not conflated.

17.6 Axiom A4 — Dimensional calibration

Statement. Every dimensional claim is accompanied by positive conversion standards and calibration maps. A minimal general package may be represented by

$$(\ell_*, \tau_*, \hbar_*).$$

Necessity rank: 2 — physical dimensions.

Countermodel on removal. Rescale

$$\ell_* \mapsto a\ell_*, \quad \tau_* \mapsto b\tau_*, \quad \hbar_* \mapsto c\hbar_*,$$

while holding every dimensionless spectral identity fixed. Absolute length, time, energy and action change. No dimensionless theorem can distinguish the rescaled models.

A4 is the smallest missing bridge from dimensionless information to dimensionful measurement. It is not produced by Shannon entropy alone.

17.7 Axiom A5 — Typed legacy-to-strict completion

Statement. There is a typed completion morphism carrying:

- amplitude;
- damping and nonlinear compression;
- phase and frequency;
- selector/topological data;
- an explicit role-semantics audit.

Necessity rank: 4 — FIN-specific completion.

Countermodel on removal. The canonical legacy and strict kernels retain the nonzero scalar-completion residual and the phase representation obstruction of Program 160.

A5 does not assert that the bridge already exists. It specifies the exact object that a conditional full FIN model must supply.

17.8 Capability theorem

All 64 subsets of $\{A_0, \dots, A_5\}$ were enumerated.

Theorem 164.1 [Proven]. The unique minimum axiom sets for the declared capabilities are:

capability	minimum axioms
dual unitary/diffusive dynamics	$\{A_0\}$
target exponent as selected state	$\{A_0, A_1\}$
generic outcome probabilities	$\{A_0, A_1, A_3\}$
signed operational outcome	$\{A_0, A_1, A_2, A_3\}$
calibrated dimensional observation	$\{A_0, A_1, A_3, A_4\}$
calibrated signed test	$\{A_0, A_1, A_2, A_3, A_4\}$
typed kernel completion	$\{A_0, A_1, A_5\}$
eligibility for full role audit	$\{A_0, \dots, A_5\}$

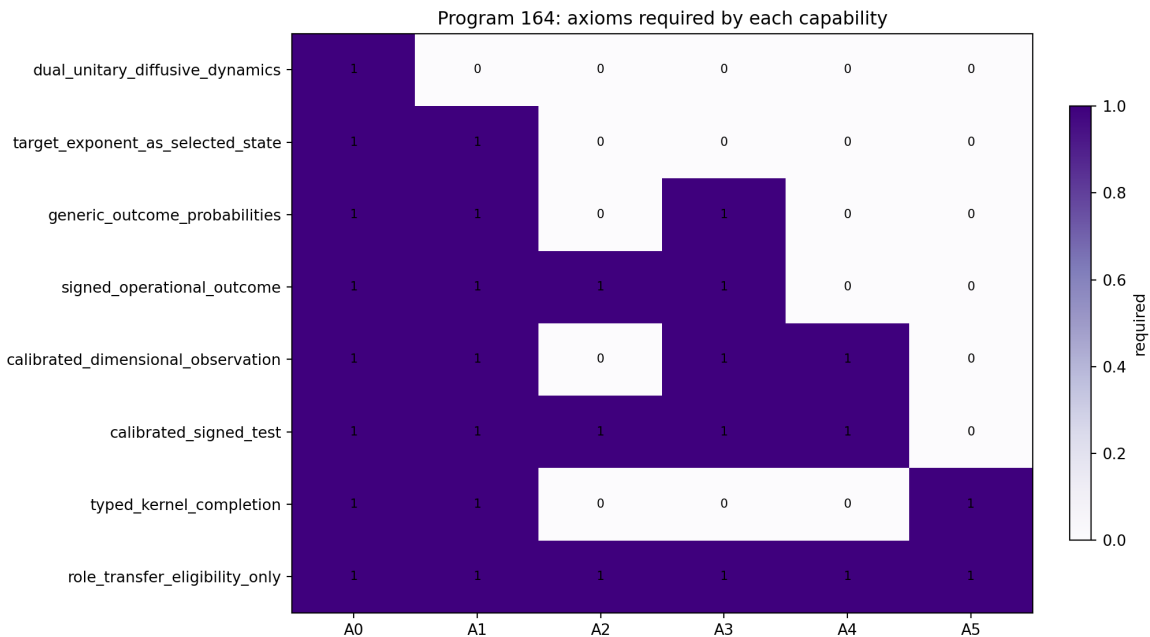


Figure 17.1: Dependency of each mathematical or physical capability on the six AFIS axioms.

17.9 Independence theorem

Theorem 164.2 [Proven]. No one of $A_1, \dots, A_5$ follows from A_0 .

Proof. Fix one conservative spectral Dirichlet generator.

1. Choose two different faithful central states: A_1 varies.
2. Choose a symmetric-only or signed preparation set: A_2 varies.
3. Choose two different instruments, or no instrument: A_3 varies.
4. Rescale the conversion standards: A_4 varies.

5. Choose no legacy-to-strict completion or a declared conditional completion: A5 varies.

All choices preserve the identical A_0 . Therefore none is a function of A_0 without extra naturality assumptions. $\square$

Corollary 164.3 [Proven]. A theorem with input only the bare generator cannot uniquely output the selected state, signed preparation, instrument, physical unit system, and FIN-specific completion map.

This is the obstruction to a single universal "missing theorem".

17.10 Smallest useful axiomatic branches

The architecture should not always deploy all six axioms.

17.10.1 Branch M — spectral mathematics

$$\{A_0\}.$$

This branch contains the spectral theorem, dual dynamics and Green family. It is the strictest mathematical core.

17.10.2 Branch S — selected dimensionless model

$$\{A_0, A_1\}.$$

With A_{ME} , it yields $\eta = 9/5$ conditionally.

17.10.3 Branch O — generic operational physics

$$\{A_0, A_1, A_3\}.$$

This branch defines outcome probabilities but need not produce a signed orientation.

17.10.4 Branch SO — signed operational physics

$$\{A_0, A_1, A_2, A_3\}.$$

This branch supports orientation-sensitive preparations and records.

17.10.5 Branch P — calibrated signed test

$$\{A_0, A_1, A_2, A_3, A_4\}.$$

This is the smallest branch able to compare a signed, dimensional prediction with a calibrated experiment.

17.10.6 Branch F — full conditional FIN

$$\{A_0, A_1, A_2, A_3, A_4, A_5\}.$$

This branch is eligible to begin a role-transfer audit. Eligibility is not proof that any legacy physical role survives.

Chapter 18

Cross-program falsification synthesis

18.1 Is one deeper object enough?

There is one deeper object for the operator constructions:

conservative spectral Dirichlet generator

It unifies:

- the spectral measure;
- wave evolution;
- diffusion;
- the Green family;
- Dirichlet energy;
- graph dynamics.

But it does not determine:

- which state is physically realized;
- which sign or branch is prepared;
- which measurement is performed;
- which dimensional scale is used;
- how legacy phase and damping complete to strict phase and compression;
- which physical interpretation is attached to a record.

The correct conclusion is therefore neither "everything is independent" nor "everything is one shadow". The repository contains a single operator core surrounded by independent operational and conversion layers.

18.2 Does the mirror coupling solve the selector?

The reflection resource theory gives a precise answer.

A mirror/reflection operation introduces a $\mathbb{Z}_2$ action. If the generator, state, preparation and instrument are all reflection-covariant, then a reflection-odd expectation remains zero. The symmetry is represented, not broken.

To break the symmetry one must add one of:

1. a signed preparation resource $M > 0$;

2. a noncovariant instrument;
3. a nonzero inversion-odd source coupled to the orientation torsor;
4. an explicit boundary or initial condition selecting a branch.

Thus a mirror term can host the missing sign, but a symmetric mirror pair does not select its own polarity. It corresponds to a missing theoretical object only when its signed coupling or preparation is independently specified.

18.3 Can information generate units?

Shannon information and $\alpha_{\text{geo}} = 4 \log 2$ are dimensionless. Any map

$$H \mapsto S_{\text{phys}}$$

requires a dimensionful conversion such as

$$S_{\text{phys}} = k_* H.$$

Similarly, a phase weight needs

$$e^{iS_{\text{phys}}/\hbar_*}.$$

The scale-rescaling countermodel proves that k_* and $\hbar_*$ cannot be uniquely recovered from dimensionless spectral identities alone. They may be axioms, measured constants, or outputs of a future scale-charged source theorem. At present they are not strict outputs.

18.4 The "observer paradox"

The same generator supports

$$U_t = e^{-itA}$$

and

$$P_t = e^{-tA}.$$

There is no mathematical contradiction: both are Borel functions of A . The apparent observer paradox begins only when one identifies one semigroup with unobserved evolution and the other with observed reduction without specifying an instrument.

A3 resolves the type error. The instrument, not P_t alone, maps a state to outcomes and post-measurement states. Diffusion may describe an environment, coarse graining, imaginary-time propagation, or a classical stochastic channel, but no one interpretation follows from functional calculus.

Chapter 19

Final theorem and final answer

19.1 Axiomatic obstruction theorem

Theorem 17.1 [Proven]. Within the category whose only input is a conservative spectral Dirichlet generator, no unique construction can simultaneously produce:

1. the target sector state;
2. a nonzero signed preparation;
3. an experimental instrument and record;
4. absolute physical dimensions;
5. a legacy-to-strict phase/compression completion.

Proof. The countermodels of Theorem 164.2 keep the input object fixed and vary each requested output independently. A unique construction from the input would have to assign the same output to all models with that input, contradicting at least one variation. $\square$

The obstruction is simultaneously:

- **state-theoretic** for the sector state;
- **resource-theoretic** for the signed preparation;
- **operational** for the instrument and record;
- **dimensional** for physical scales;
- **representation-theoretic** for the phase completion.

It is not merely algebraic.

19.2 Deepest surviving interpretation

After the attempted constructions and falsifications, the deepest interpretation is:

FIN is most coherently formulated as an axiomatic informational-spectral operational theory. Its strict mathematical core is a conservative spectral Dirichlet generator associated with the informational nadsoliton. The generator unifies Green response, graph energy, unitary propagation and diffusive propagation. A physical theory arises only after independent state, preparation, instrument and calibration structures are supplied. The legacy-to-strict completion is a further FIN-specific obligation. No current theorem collapses these layers into one unavoidable object.

The most promising new axiom is A_{ME} because it closes one clearly identified problem with one

equation and has a uniqueness proof. Its proper status is **conditional**, and that status is a scientific strength rather than a defect: the premise can now be attacked directly.

Chapter 20

Recommended Programs 165–177

The next studies should test the axioms instead of adding them silently.

20.1 Program 165 — Arb-certified oscillatory tail splitting

Replace absolute tail domination by residue-class or Fourier-mode cancellation, use Arb/acb ball arithmetic, and certify the full $[10^{-3}, 2 \cdot 10^{-2}]$ window.

Probability of a materially tighter theorem: 0.80.

Probability of a sub-3% theorem with local resources: 0.45.

20.2 Program 166 — Derive or falsify A_{ME} compositionally

Ask whether tensor additivity, independent composition of fibre systems, real-character functoriality, and stability under coarse graining force

$$\beta_F = \alpha_{\text{geo}}/4.$$

Construct counterexamples before seeking uniqueness.

Probability of a useful classification/no-go: 0.85.

Probability of strict derivation: 0.20.

20.3 Program 167 — Formal AFIS consistency and independence

Encode the finite-dimensional AFIS model and all removal countermodels in Lean, Isabelle, or another proof assistant. Separate analytic Dirichlet-form assumptions from finite matrix witnesses.

Probability of formal completion: 0.80.

20.4 Program 168 — Effective all-scale irrationality measure

Search for an explicit theorem strong enough to establish $D(\kappa, \nu)$ for $743/(4000\pi)$, then propagate its actual constants into the Abelian remainder. Stop if constants are mathematically valid but numerically useless.

Probability of some effective bound: 0.55.

Probability of a useful practical constant: 0.15.

20.5 Program 169 — Monoidal groupoid valuation theorem

Extend Program 153 from invariance to tensor products, induction, restriction, and valuation additivity. Classify all compatible orbit masses.

Probability of classification: 0.80.

Probability of uniquely selecting uniform sector mass: 0.25.

20.6 Program 170 — Stability of the A_{ME} equilibrium

Perturb dimensions, energy gaps, degeneracies and β_F . Derive the condition number of η and determine whether $9/5$ is structurally stable or fine-tuned.

Probability of a complete perturbation theorem: 0.95.

20.7 Program 171 — Full reflection-resource theory

Use semidefinite programming and representation theory to classify reflection-covariant channels on the full density space. Determine whether a finite family of monotones replaces M .

Probability of strong finite results: 0.85.

20.8 Program 172 — Nonasymptotic detector statistics

Add correlated noise, pixelization, censoring, estimated detector response and finite-memory channels. Derive honest bootstrap or concentration intervals and compare IQR with empirical characteristic-function estimators.

Probability of an operationally useful procedure: 0.85.

20.9 Program 173 — Calibration-control experimental design

Construct the smallest set of known-input controls that makes the exponent score transverse to the apparatus tangent space. Optimize measurement times and allocation under a fixed record budget.

Probability of an identifiability theorem: 0.90.

20.10 Program 174 — Composite adversarial preregistration

Replace the binary Program-150 rule with a preregistered comparison among Gaussian, stable, tempered, truncated, correlated, and gain-confounded families. Include model checking and abstention.

Probability of reducing false FIN attribution: 0.95.

20.11 Program 175 — General categorical phase obstruction

Classify all equivariant maps among the legacy $\mathbb{Z}_8$ character, the $\mathbb{Z}_{12}$ carrier and the strict infinite-order phase. Quantify the minimal symmetry-breaking data required by a nonlinear completion.

Probability of a general obstruction theorem: 0.85.

20.12 Program 176 — One explicit nonlinear completion candidate

Construct exactly one typed nonlinear map with independent amplitude, compression and phase-source slots. Audit it against the six Program-162 edges. Stop at the first failed edge.

Probability of a useful obstruction: 0.80.

Probability of complete positive bridge: 0.10.

20.13 Program 177 — AFIS measurement model for double-slit records

Build a finite CP-instrument model with preparation, propagation, environmental decoherence, detector resolution and persistent record. Demonstrate exactly which interference and which diffusion statements follow from A_0 – A_4 , without attributing collapse to P_t alone.

Probability of a rigorous conditional model: 0.90.

Probability of strict physical validation without data: 0.

20.14 Priority

The recommended order is

$$166 \rightarrow 170 \rightarrow 167 \rightarrow 173 \rightarrow 174 \rightarrow 172 \rightarrow 165 \rightarrow 171 \rightarrow 175 \rightarrow 177 \rightarrow 169 \rightarrow 168 \rightarrow 176.$$

The priority is intentionally axiom-first:

1. try to derive or falsify the most economical new axiom;
 2. quantify its stability;
 3. formalize independence;
 4. repair experimental identifiability;
 5. return to the expensive bridge only with a typed candidate.
-

Chapter 21

Reproducibility statement

The executable research file is:

`fin_programs_151_164_axiomatic_operational_foundations.py`

The machine-readable results are:

`FIN_Programs_151_164_Axiomatic_Operational_Results.json`

The extracted axiom system is:

`FIN_Programs_151_164_Minimal_Axiomatic_System.json`

Regression tests are:

`test_fin_programs_151_164_axiomatic_operational_foundations.py`

Fourteen figures are generated in:

`FIN_Programs_151_164_Axiomatic_Operational_Figures/`

The study uses no external physical data.

Chapter 22

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Chapter 23

Suggested citation

Żuchowski, K. (2026). *FIN Programs 151–164: Axiomatic Operational Foundations, State Selection, and Falsifiable Measurement* (FIN Research Monograph, Release 10.15; Version 1.0.0) [Preprint]. Zenodo.

Archival note

Machine-readable results are stored in `FIN_Programs_151_164_Axiomatic_Operational_Results.json`. The extracted six-axiom system is stored in `FIN_Programs_151_164_Minimal_Axiomatic_System.json`. The executable, sixty-two tests, and fourteen figures reproduce the release.

Suggested citation:

Żuchowski, K. (2026). *FIN Programs 151–164: Axiomatic Operational Foundations, State Selection, and Falsifiable Measurement* (FIN Research Monograph, Release 10.15; Version 1.0.0) [Preprint]. Zenodo.